Name: Amoroso, Ivan Karl J.
Section: 3.4 BSIT

Course: IT-Elective 2
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Activity 2: Library Book Management

Code:

Program.cs

LibraryApp

Book

```
using System;

// Book Class represents a physical book in the library
8 references
class Book
{
    7 references
    public string Title { get; set; }
    3 references
    public string Author { get; set; }
    3 references
    public string Isbn { get; set; }
    7 references
    public bool IsAvailable { get; private set; }

    3 references
    public Book(string title, string author, string isbn)
    {
        Title = title;
        Author = author;
        Isbn = isbn;
        IsAvailable = true;
    }

    // Method to borrow the book
    2 references
    public void Borrow()
    {
        if (IsAvailable)
        {
            IsAvailable = false;
            Console.WriteLine($"\"{Title}\" has been borrowed.");
        }
        else
        {
            Console.WriteLine($"Sorry, '{Title}' is currently not available.");
        }
    }

```

Program.cs

LibraryApp | Program | Main(string[] args)

```
33      // Method to return the book
        1 reference
34      public void Return()
35      {
36          if (!IsAvailable)
37          {
38              IsAvailable = true;
39              Console.WriteLine($"\"{Title}\" has been returned.");
40          }
41          else
42          {
43              Console.WriteLine($"\"{Title}\" was not borrowed.");
44          }
45      }
46
47      // Method to check out the book
        6 references
48      public virtual void DisplayInfo()
49      {
50          Console.WriteLine($"Title: {Title} | Author: {Author} | ISBN: {Isbn} | Available: {IsAvailable}");
51      }
52  }
53
54  // EBook Class represents an electronic book in the library
    4 references
55  class EBook : Book
56  {
        2 references
57      public double FileSizeMb { get; set; }
        2 references
58      public EBook(string title, string author, string isbn, double fileSizeMb)
59          : base(title, author, isbn)
60      {
61          FileSizeMb = fileSizeMb;
62      }
63
64      // Override the DisplayInfo method to include file size
        6 references
65      public override void DisplayInfo()
66      {
67          Console.WriteLine($"Title: {Title} | Author: {Author} | ISBN: {Isbn} | Available: {IsAvailable} | File Size (MB): {FileSizeMb}");
68      }
69  }
```

Program.cs

LibraryApp | Program

```
70
71  // Main Class Program
    0 references
72  class Program
73  {
        0 references
74      static void Main(string[] args)
75      {
76          // Creating 2 Book objects and 1 EBook object
77          Book book1 = new Book("Python Basics", "Ivan Amoroso", "111");
78          Book book2 = new Book("JavaScript Basics", "Erik Trautman", "222");
79          EBook ebook1 = new EBook("Programming Fundamentals", "David Malan", "333", 1.5);
80
81          // Displaying information of the books
82          Console.WriteLine("Library Books Information:");
83          book1.DisplayInfo();
84          book2.DisplayInfo();
85          ebook1.DisplayInfo();
86
87          // Borrowing a book
88          Console.WriteLine("\nBorrowing a book...");
89          book1.Borrow(); // Borrowing the first book
90          book1.Borrow(); // Attempt to borrow again
91          book1.Return(); // Returning the book
92          book1.DisplayInfo(); // Display updated availability
93
94          // Polymorphism demonstration
95          Console.WriteLine("\nPolymorphism Demonstration:");
96          Book libraryItem = new EBook("Data Structures", "Rene Decartes", "444", 2.0);
97          libraryItem.DisplayInfo();
98      }
99  }
```

Output:

Microsoft Visual Studio Debu

```
Library Books Information:
Title: Python Basics | Author: Ivan Amoroso | ISBN: 111 | Available: True
Title: JavaScript Basics | Author: Erik Trautman | ISBN: 222 | Available: True
Title: Programming Fundamentals | Author: David Malan | ISBN: 333 | Available: True | File Size (MB): 1.5

Borrowing a book...
"Python Basics" has been borrowed.
Sorry, 'Python Basics' is currently not available.
"Python Basics" has been returned.
Title: Python Basics | Author: Ivan Amoroso | ISBN: 111 | Available: True

Polymorphism Demonstration:
Title: Data Structures | Author: Rene Decartes | ISBN: 444 | Available: True | File Size (MB): 2
```